

BST Physics Curriculum Vol 6 Chapter 6 — Classical Statistical Mechanics v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready; refilled per Cal #104)

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Vol 6 Chapter 6 — Classical Statistical Mechanics

Chapter motivation

Standard classical statistical mechanics derives macroscopic thermodynamics from microscopic statistics. The fundamental distributions — Boltzmann $P(E) \propto e^{-\beta E}$, Maxwell-Boltzmann velocity distribution $f(v) \propto v^2 e^{-mv^2/(2k_B T)}$, equipartition theorem $\langle \frac{1}{2}mv^2 \rangle = \frac{1}{2}k_B T$ per quadratic degree of freedom — emerge from counting microstates at fixed macroscopic constraints (energy, volume, particle number). Standard texts: Pathria + Reif + Landau-Lifshitz Vol 5.

BST identifies classical stat mech as substrate-equilibrium emergent at coarse-grained limit: standard distributions inherit from substrate partition function $Z = \text{Tr}(e^{-\beta H_{\text{sub}}})$ (Vol 6 Ch 5 LOAD-BEARING) in the limit where quantum coherence is negligible at relevant temperature. The substrate-tick Casimir spectrum becomes effectively continuous; substrate Boltzmann weight matches standard Boltzmann distribution.

Section 6.0b — Reader-grade 3-level pedagogy

Level 1: Standard Boltzmann + Maxwell-Boltzmann distributions emerge as macroscopic limit of substrate heat-kernel evaluation (Vol 6 Ch 5) when classical-physics approximation valid (no quantum coherence at relevant scale); equipartition theorem $\langle \frac{1}{2}mv^2 \rangle = \frac{1}{2}k_B T$ inherits from substrate Casimir eigenvalue spread per quadratic degree of freedom.

Level 2 (graduate-physicist): Classical Boltzmann distribution $P(E) \propto e^{-E/k_B T}$ derives from substrate partition function $Z = \text{Tr}(e^{-\beta \text{Casimir}})$ in classical limit (Casimir spectrum continuous + quantum coherence negligible at macroscopic

temperature). Maxwell-Boltzmann velocity distribution $f(v) \propto v^2 e^{-mv^2/(2k_B T)}$ emerges from substrate momentum K-type distribution at thermal equilibrium with Casimir eigenvalues approximated as continuous. Equipartition theorem $\langle \frac{1}{2}mv^2 \rangle = \frac{1}{2}k_B T$ per quadratic degree of freedom from substrate Casimir eigenvalue spread $\times$ substrate-tick computational rate (Vol 6 Ch 1 temperature reading). Gibbs ensembles (microcanonical, canonical, grand canonical) are substrate-zone-2 commitment marginalizations at fixed conserved quantities: microcanonical = fixed energy + volume + particle number; canonical = fixed temperature (substrate-tick rate); grand canonical = fixed chemical potential (substrate K-type chemical-equivalent). Standard equation of state (ideal gas $PV = Nk_B T$) recovered from substrate equipartition + classical-limit pressure substrate-cartography.

Level 3 (5th-grader accessible): Classical statistical mechanics describes systems with many particles (gases, liquids) using probability distributions. Boltzmann's distribution says $P(E) \propto e^{-E/k_B T}$ (high-energy states are exponentially less likely than low-energy states). Maxwell-Boltzmann gives the velocity distribution for gas molecules. Equipartition says each "degree of freedom" gets $\frac{1}{2}k_B T$ of energy on average. BST recovers these as the macroscopic limit of substrate heat-kernel evaluation (Vol 6 Ch 5). Quantum corrections matter only at low temperature or small systems — at room temperature for everyday gases, classical statistical mechanics works perfectly.

Section 6.1 — Standard Classical Distributions

Boltzmann distribution: $P(E) \propto e^{-E/k_B T}$ for energy E at temperature T .

Maxwell-Boltzmann velocity distribution: $f(v) d^3v \propto v^2 e^{-mv^2/(2k_B T)} d^3v$ for ideal gas molecules of mass m .

Equipartition theorem: each quadratic degree of freedom contributes $\frac{1}{2}k_B T$ to average energy $\langle E \rangle$.

Ideal gas equation of state: $PV = Nk_B T$ from equipartition + kinetic theory.

Section 6.2 — Substrate-Cartography Reading

Standard Boltzmann distribution $P(E) \propto e^{-E/k_B T}$ = substrate partition function evaluation in classical limit:

$$Z(\beta) = \text{Tr}(e^{-\beta \text{Casimir}}) \rightarrow \int e^{-\beta E} \rho(E) dE \text{ in continuous-spectrum limit}$$

where $\rho(E)$ is the density of Casimir eigenstates per substrate-tick. Macroscopic temperature $T = \text{substrate-tick computational rate} \times \text{Casimir-eigenvalue spread}$ (Vol 6 Ch 1 substrate dictionary).

Maxwell-Boltzmann velocity distribution $f(v)$: substrate momentum K-type $V_{-}(p,q)$ distribution at thermal equilibrium with Casimir eigenvalues approximated as con-

tinuous. The v^2 factor in $f(v)$ comes from substrate momentum-K-type density at fixed $|p|$.

Section 6.3 — Gibbs Ensembles Substrate-Bookkeeping

Microcanonical ensemble: fixed $E + V + N$. Substrate: fixed Zone-2 commitment K-type subset coverage; equal probability over all microstates consistent with macroscopic constraints.

Canonical ensemble: fixed $T + V + N$. Substrate: fixed substrate-tick rate (temperature reading) + Zone-2 commitment + fixed particle K-type count.

Grand canonical ensemble: fixed $T + V + \mu$. Substrate: chemical potential μ corresponds to substrate K-type creation/annihilation cost.

Section 6.4 — Equipartition + Ideal Gas Recovery

Equipartition theorem $\langle \frac{1}{2}mv^2 \rangle = \frac{1}{2}k_B T$ per quadratic degree of freedom derives from substrate Casimir eigenvalue spread per substrate-tick $\times$ Boltzmann weight integration:

$$\langle E \rangle = \int E \cdot P(E) dE = \frac{1}{2}k_B T \text{ per quadratic DOF}$$

Ideal gas $PV = Nk_B T$ recovered from equipartition + standard kinetic-theory pressure derivation. Substrate-cartography: pressure = substrate-tick boundary-force rate (Vol 6 Ch 1).

Section 6.5 — Honest scope + Connection

What this chapter establishes: classical stat mech distributions + ensembles + equipartition + ideal gas all derive from substrate partition function (Vol 6 Ch 5 LOAD-BEARING) in classical limit. No new postulates beyond substrate framework.

Falsifier: any system where classical Boltzmann distribution fails at high-T classical-limit conditions falsifies substrate-derivation. Standard QM corrections expected at low-T (Bose-Einstein + Fermi-Dirac per Vol 6 Ch 7).

Open scope (multi-week): - Explicit $\rho(E)$ density-of-states derivation from substrate Casimir spectrum - Equipartition factor ($\frac{1}{2}$) from substrate-symmetry constraint precise derivation - Standard QM-classical crossover temperature substrate-mechanism

Connection: - Vol 6 Ch 1 substrate thermodynamic dictionary - Vol 6 Ch 5 LOAD-BEARING partition function = heat kernel - Vol 6 Ch 7 Quantum stat mech (Bose-Einstein + Fermi-Dirac substrate-derivations) - Vol 5 Ch 4 Schrödinger equation (Hamiltonian framework cross-link)

— Lyra, Vol 6 Ch 6 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT